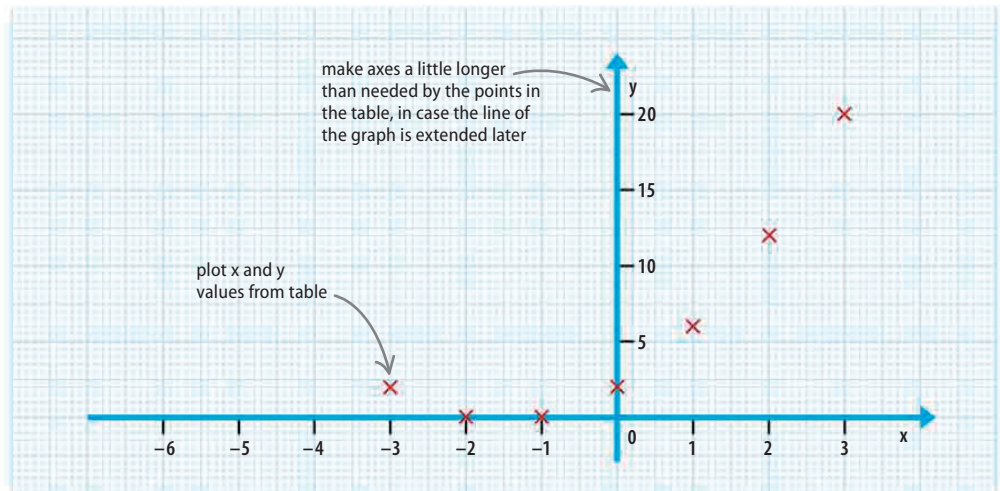


Draw the graph of the equation. Use the values of x and y that have been found in the table as the coordinates of points on the graph. For example, $x = 1$ has the corresponding value $y = 6$. This becomes the point on the graph with the coordinates $(1, 6)$.

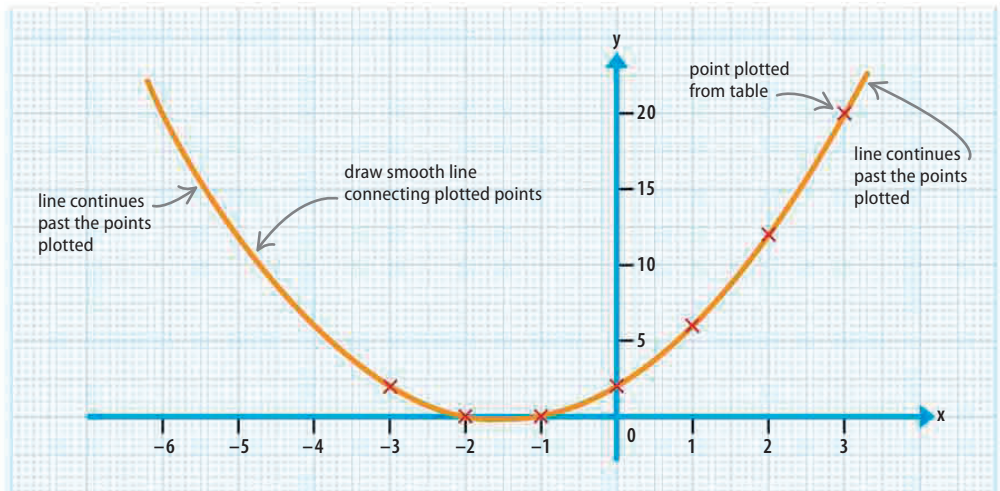
▷ Draw the axes and plot the points

Draw the axes of the graph so that they cover the values found in the tables. It is often useful to make the axes a bit longer than needed, in case extra values are added later. Then plot the corresponding values of x and y as points on the graph.



▷ Join the points

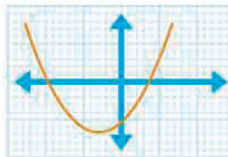
Draw a smooth line to join the points plotted on the graph. This line is the graph of the equation $y = x^2 + 3x + 2$. Bigger and smaller values of x could have been chosen, so the line continues past the values that have been plotted.



LOOKING CLOSER

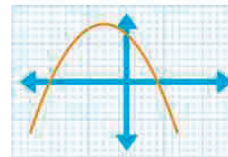
The shape of a quadratic graph

The shape of a quadratic graph depends on whether the number that multiplies x^2 is positive or negative. If it is positive, the graph is a smile; if it is negative, the graph is a frown.



$$\triangleleft y = ax^2 + bx + c$$

If the value of the a term is positive, then the graph of the equation is shaped like this.



$$\triangleleft y = -ax^2 + bx + c$$

If the value of the a term is negative, then the graph of the equation is shaped like this.